ARI BLOCH

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EDUCATION

University of Wisconsin-Madison Madison, WI

Bachelor of Science in Aerospace Engineering and Mathematics | GPA: 3.2/4.0 Expected May 2028

Relevant Coursework: Aerodynamics, Gas Dynamics, Intermediate Fluid Dynamics (grad-level), Aerospace Structures, Finite Element Analysis, Numerical Analysis, Orbital Mechanics, Advanced Dynamics, Partial Differential Equations

Organizations: American Institute of Aeronautics and Astronautics (AIAA), American Society of Mechanical Engineers (ASME), WiscWind, Acacia Fraternity

EXPERIENCE

Undergraduate Researcher Sep 2025 - Present

Computational Flow Physics and Modeling Group, Dept. of Mechanical Engineering, UW-Madison Madison, WI

- Simulate vertical-axis cross-flow turbine hydrodynamics in OpenFOAM (k-omega SST, RANS, sliding mesh) to study virtual camber effects on a rotating NACA 0018 blade at a Reynolds number of 1 million.
- Ran a 12-case structured mesh refinement study in Pointwise, varying first-cell height, boundary-layer growth, and cell aspect ratio; tracked down unstable force results to a potential turbulence parameter mismatch.
- Automate parallel CFD runs and post-processing on the CHTC HPC cluster using Linux, Bash, and SLURM batch scripts for domain decomposition, field reconstruction, and data archiving.
- Develop Python mesh-morphing workflows to add camber without re-meshing; built a Python tool that visualizes virtual camber over the rotation cycle.
- Developed a physics-based heat transfer model of ice-covered lakes in Python, pairing an analytical radiation-damming solution with a finite-difference solver and Beer-Lambert solar forcing.
- Built for Lake Mendota algae-transport research; presented at the 2026 UW-Madison Undergraduate Symposium.

Practice Engineering Problem Solving (PrEPS) Facilitator Jan 2026 - Present

College of Engineering Undergraduate Learning Center, UW-Madison Madison, WI

- Lead weekly sessions for 15-20 Calculus II students, designing lesson plans around commonly missed exam concepts.

Mathematics & Fluid Dynamics Tutor Sep 2025 - Present

- Tutor Calculus I-III, Linear Algebra, Differential Equations, and Fluid Dynamics one-on-one.

Outreach Team Member Oct 2026 - Present

AIAA Student Chapter, UW-Madison Madison, WI

- Helping re-establish UW-Madison's inactive student chapter of American Institute of Aeronautics and Astronautics.
- Recruit faculty and research-lab speakers for a weekly technical speaker series.

PROJECTS

Wind Tunnel Airfoil Design and Testing | Intro to Aerospace Engineering Sep 2024 - Dec 2024

- Designed and fabricated airfoils in SolidWorks and tested them in an open-loop wind tunnel, validating measured lift and drag against QBlade predictions.
- Wrote a MATLAB tool generating cambered NACA four-digit airfoils for rapid screening across angles of attack.
- Supplied QBlade load data to partner teams for structural analysis; helped test a servo-actuated angle-of-attack mount.

SKILLS

CAE: OpenFOAM, ANSYS Mechanical APDL, ANSYS Workbench, Pointwise, QBlade, Simulink
CAD: SolidWorks, Siemens Inspire
Programming & Computing: Python, MATLAB, Linux, Bash, SLURM, HPC cluster computing (CHTC), Microsoft Excel
Engineering Methods: Computational fluid dynamics (CFD), Unsteady RANS turbulence modeling (k-omega SST), mesh generation, aerodynamic analysis, wind tunnel testing, finite element analysis (FEA), numerical methods